ASSESSMENT OF PREDISPOSING FACTORS TO TREATMENT FAILURE AMONG EXTENSIVELY DRUG-RESISTANT TUBERCULOSIS PATIENTS ATTENDING TB CLINICS IN KWARA STATE

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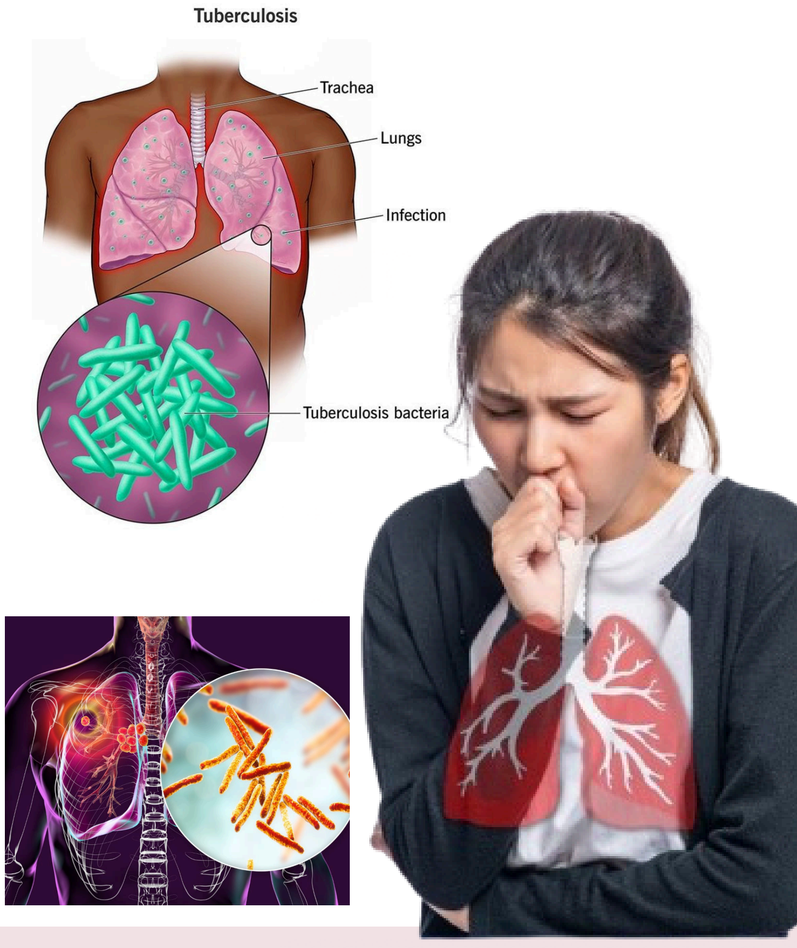
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INTRODUCTION

TUBERCULOSIS (TB) IS A CONTAGIOUS AND POTENTIALLY LIFE-THREATENING DISEASE CAUSED BY MYCOBACTERIUM TUBERCULOSIS, PRIMARILY AFFECTING THE LUNGS BUT CAPABLE OF SPREADING TO OTHER ORGANS.

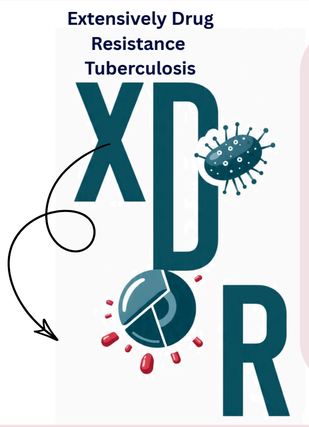
Despite the availability of effective anti-TB drugs, TB remains one of the leading causes of death from infectious diseases globally, particularly in developing countries such as Nigeria (World Health Organization, 2025).

- Extensively drug-resistant tuberculosis (XDR-TB) is a severe form of TB characterized by resistance to Rifampicin and Isoniazid, as well as Fluoroquinolones and at least one second-line injectable drug(Kanamycin,Capreomycin) (WHO, 2023).
- A major driver of this resistance is incomplete or inadequate treatment, where poor adherence exposes bacteria to sub-therapeutic drug levels, promoting the emergence of resistant strains (*Lungu et al., 2022*).
- XDR-TB presents serious public health challenges, including prolonged treatment duration, increased toxicity, higher treatment failure rates, and increased mortality (*Adepoju et al., 2021*).
- Additional contributing factors include HIV co-infection, poor drug quality, lifestyle factors, and adverse socio-economic conditions.
- Understanding these factors is essential for improving treatment adherence, strengthening TB control programs, and guiding effective public health interventions.



STATEMENT OF PROBLEM

Tuberculosis (TB) remains a major public-health challenge, particularly in developing countries like Nigeria, where it continues to cause significant illness and death despite the availability of effective treatment. The increasing emergence of drug-resistant forms of TB, especially extensively drug-resistant tuberculosis (XDR-TB), has made disease control more difficult. These resistant strains are more difficult and costly to treat, often resulting in prolonged illness, poor treatment outcomes, and increased risk of death.



AIM

To determine the predisposing factors responsible for the incomplete or interrupted treatment of anti-TB treatment.

OBJECTIVES

- To determine the prevalence of extensively drug-resistant tuberculosis (XDR-TB) among previously treated tuberculosis patients.
- To identify predisposing factors responsible for incomplete or interrupted anti-tuberculosis treatment.
- To find the association between the factors responsible for the incomplete anti-tuberculosis treatment and emergence of XDR TB.

METHODOLOGY

STUDY DESIGN : Descriptive cross-sectional hospital based study.

STUDY POPULATION: Patients on tuberculosis treatment attending the TB Clinics in Kwara State.

SAMPLE SIZE DETERMINATION:

- Based on an expected prevalence rate of 19.18% XDR -TB in North Central, Nigeria (Oladimeji et al., 2022)
- The minimum sample size would be approximately 238 participants.

STUDY AREA: TB Clinics in Kwara State.

DATA COLLECTION: Structured Questionnaires

DATA ANALYSIS: Statistical analysis using (e.g SPSS)

EXPECTED OUTCOME

It is expected that this study will reveal the proportion of extensively drug-resistant tuberculosis (XDR-TB) among patients with incomplete tuberculosis treatment and establish a clear association between incomplete therapy and the development of drug resistance. The findings will provide valuable insight into the importance of treatment adherence and help in strengthening tuberculosis control strategies. It is also anticipated that the study outcomes will guide healthcare professionals and policymakers in implementing more effective monitoring and follow-up systems to reduce the burden of drug-resistant TB in Nigeria.

ALIGNMENT WITH SDGs

This study contributes directly to SDG 3: Good Health and Well-Being, particularly target 3.3, which aims to end the epidemics of tuberculosis and other communicable diseases by 2030. By assessing factors that lead to treatment failure among XDR-TB patients, the study will provide evidence to strengthen TB control programs, improve treatment adherence, and reduce the spread of resistant TB strains.

CONCLUSION

Treatment failure in extensively drug-resistant tuberculosis (XDR-TB) is influenced by multiple factors including patient adherence, clinical conditions, and socio-economic challenges.Identifying and addressing these factors is essential for improving treatment outcomes and reducing disease transmission.

Strengthening diagnostic, monitoring, and patient support systems will enhance the effectiveness of TB control programs.

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